

AWS In – ServiceHub (Web/Mobile)

Project: [Full-Stack Local Services Platform (ServiceHub)]

A secure, scalable, and high-performance application connecting users with verified local service providers (plumbers, electricians, carpenters, etc.), enabling location-based discovery, real-time booking, and trusted interactions.

Introduction

ServiceHub aims to revolutionize the way users discover and book local services. By using **Amazon Web Services (AWS)**, the platform ensures scalability, security, and reliability. AWS provides the infrastructure to handle dynamic workloads, protect sensitive data, and deliver seamless experiences across web and mobile applications.

Amazon EC2(Elastic Compute Cloud)

- Manage **backend APIs** and **databases** for ServiceHub.
- Auto Scaling ensures the platform can handle peak booking times without downtime.
- Example: Multiple EC2 instances running Node.js backend with MySQL for service provider and booking data.

AWS Lambda

- Executes **event-driven tasks** without server management.
- Use cases:
 - Sending booking confirmations via email/SMS.
 - Sending notifications when providers accept jobs.
 - Processing payments asynchronously.
- Reduces operational costs by charging only for execution time.

Amazon S3(Simple Storage Service)

- Stores user and provider documents (ID proofs, certifications).
- Hosts service-related images (before/after work photos).
- Store payment- receipts or invoices of users and service-provider.
- Provides secure, durable storage with versioning and lifecycle policies.
- Can integrate with CloudFront for fast content delivery.

IAM and Security

- Identity and Access Management (IAM) using OTP verification.
- Implements **role-based access control**:
 - Admins: Full system access.
 - Providers: Limited access to booking and profile data.
 - Users: Restricted to personal accounts and bookings.
- Multi-Factor Authentication (MFA) ensures secure logins.
- Encryption protects sensitive data like addresses and payment details.

Load Balancer and Reliability

- Elastic Load Balancing distributes traffic across EC2 instances.
- Ensures **high availability** and fault tolerance.
- Combined with Auto Scaling, ServiceHub remains responsive during traffic surges (e.g., festival seasons such as Diwali, when families often book cleaning, painting, and home decoration services).
- Improves user trust by minimizing downtime.